ANSWERS

1	a	2	b	3	c	4	b	5	c	6	a	7	c	8	c	9	a	10	a
11	a	12	a	13	d	14	c	15	c	16	a	17	b	18	c	19	a	20	d
21	c	22	a	23	a	24	c	25	c	26	c	27	a	28	c	29	c	30	c
31	c	32	b	33	d	34	a	35	a	36	d	37	d	38	b	39	a	40	a
41	d	42	c	43	c	44	c	45	c	46	c	47	a	48	a	49	d	50	c
51	a	52	b	53	a	54	a	55	d	56	d	57	c	58	b	59	b	60	c
61	c	62	c	63	d	64	a												

HINTS AND SOLUTIONS

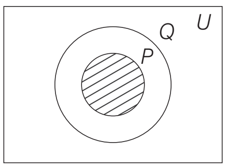
1. *(a)* $A \cup B = \{5, 6, 7\} \cup \{7, 8, 9\}$
$= \{5, 6, 7, 8, 9\}$

2. *(b)* $B - A = \{7, 8, 9, 12\} - \{2, 6, 8, 9\}$
$= \{7, 12\}$

3. *(c)* Given, $A = \{1, 2, 3, 4, 5\}$ and $A \subseteq U$.
$\Rightarrow A \cap U = \{1, 2, 3, 4, 5\} \cap \{U\}$
$= \{1, 2, 3, 4, 5\}$

4. *(b)* Let $A = \{2, 4, 16, 256, ...\}$
for $n = 0$, $2^{2^0} = 2^1 = 2$
for $n = 1$, $2^{2^1} = 2^2 = 4$
for $n = 2$, $2^{2^2} = 2^4 = 16$
Thus,
$A = \{x \in N \mid x = 2^{2^n},\ n = 0, 1, 2, ...\}$

5. *(c)* $\because\ P \subset Q$



So, $P \cap Q = P$

6. *(a)* Let $x \in A - B \Rightarrow x \in A$ and $x \notin B$
$\Rightarrow x \notin B - A$
Also, $x \in B - A \Rightarrow x \in B$
and $x \notin A \Rightarrow x \notin A - B$
So, $(A - B) \cap (B - A) = \phi$

7. *(c)* Only if P and Q both are empty set or P and Q have same elements, then $P \cup Q = P \cap Q$.

8. *(c)* Number of elements in $A = 4$
$\therefore$ Total number of non-empty
Proper subsets of $A = 2^n - 2$
$= 2^4 - 2 = 14$

9. *(a)* Given, $A = \{B, O, W, L\}$
$B = \{B, O, W, L, E\}$
$C = \{B, O, W, L, E\}$
$\therefore\ A \subset B$ and $B = C$

10. *(a)* Clearly, $A \cap (A \cup B) = A$

11. *(a)* $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$
$\Rightarrow B = \{3, 5, 9\}$ is the required smallest set

12. *(a)* Here, $(P \cup R) = \{0, 1, 2, 3, 4, 5, 6\}$
$Q' = U - Q = \{0, 1, 6, 7, 8, 9\}$
So, $Q' \cap (P \cup R) = \{0, 1, 6\}$

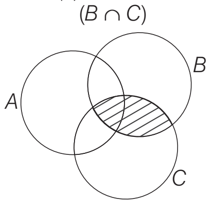
13. *(d)* Let $U = \{1, 2, 3, 4, 5, 6\}$ and $P = \{1, 3, 4\}$
Then, $P' = U - P = \{2, 5, 6\}$
and $(P')' = U - P' = \{1, 3, 4\} = P$

14. *(c)* The shaded region is $U - A = A'$.

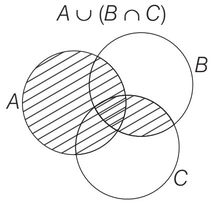
15. *(c)* The shaded region represents the elements in A and not in B, so it represents $A - B$.

16. *(a)* We have,

$(B \cap C)$



$A \cup (B \cap C)$



17. *(b)* The shaded region represents the elements in A and not in B or C.
$\therefore\ A - (B \cup C)$

18. *(c)* Clearly, $(A \cap B) \cup (B \cap C)$

19. *(a)* Clearly, (a) is correct.

20. *(d)* Since, $P(\phi) = \{\phi\}$
Hence, $\phi \in P(\phi)$

21. *(c)* As, $\pi, \sqrt{2}, 3 + \sqrt{7}$ are irrational numbers.
So, required set is $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$.

22. *(a)* Here, $U = \{1, 2, 3, 4, 5, ...\}$
and $A = \{1, 3, 5, 7, 9, 11, \ldots\}$
So, $A' = U - A = \{2, 4, 6, 8, 10, \ldots\}$
$\therefore\ A' = \{x : x$ is an even number$\}$

23. *(a)* Disjoint sets have no common element.
e.g. $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$
$\Rightarrow\ A \cap B = \phi$

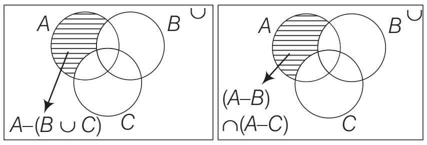
24. *(c)* $A - (A - B) = A - (A \cap B')$
$= A \cap (A \cap B')' = A \cap (A' \cup B)$
$= (A \cap A') \cup (A \cap B)$
$= \phi \cup (A \cap B) = A \cap B$

25. *(c)* $\because\ n(A \cap B) = 2n$,
$\therefore\ n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 2n + 4n - 2n = 4n$
Hence, minimum number of elements in $A \cup B$ is $4n$.

26. *(c)* Here, $P = \{x : x^2 - 3x + 2 = 0\}$
$\Rightarrow\ P = \{x : (x - 1)(x - 2) = 0\}$
$\Rightarrow\ P = \{x : x = 1, x = 2\}$
$\therefore\ P = \{1, 2\}$ and
$Q = \{x : x^2 + 4x - 12 = 0\}$
$\Rightarrow\ Q = \{x : (x + 6)(x - 2) = 0\}$
$\Rightarrow\ Q = \{-6, 2\}$
So, $P - Q = \{1, 2\} - \{-6, 2\} = \{1\}$

27. *(a)* Given, $A = \{(2^{2n} - 3n - 1) \mid n \in N\}$
$= \{0, 9, 54, 243, \ldots\}$
and $B = \{9(n - 1) \mid n \in N\}$
$= \{0, 9, 18, 27, \ldots\}$
From above, it is clear that $A \subset B$.

28. *(c)*



From the above two figures, shaded portion of
$A-(B\cup C)=(A-B)\cap(A-C)$

29. *(c)* $n(P\cup Q)=n(P)+n(Q)-n(P\cap Q)$
$\therefore \quad n(P\cup Q)=m+n-p$

30. *(c)* $n(A\cup B)=n(A)+n(B)-n(A\cap B)$
$\therefore n(A\cap B)=n(A)+n(B)-n(A\cup B)=17+23-38=2$

31. *(c)* Given, $A=\{x: x^2-6x+8=0\}$
$\Rightarrow \quad A=\{x:(x-4)(x-2)\}$
$\therefore \quad A=\{4, 2\}$
And $\quad B=\{x: 2x^2+3x-2=0\}$
$\Rightarrow \quad B=\{x:(2x-1)(x+2)\}$
$\therefore \quad B=\left\{\frac{1}{2},-2\right\}$
Hence, neither $A\subseteq B$ nor $B\subseteq A$.

32. *(b)* $(B\cup C)=\{1, 2, 3, 4, 5, 6, 7\}$
$\therefore \quad A\cap(B\cup C)=\{1,2,3,4\}\cap\{1,2,3,4,5,6,7\}=\{1,2,3,4\}$

33. *(d)* $(A\cup B)=\{1,2,3,4,6,8,10\}$
$(A\cup B)'=U-(A\cup B)$
$=\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}-\{1, 2, 3, 4, 6, 8, 10\}$
$=\{5,7,9\}$

34. *(a)* $\{(A\cup B)'\cap A\}-(A-B)$
$=\{(U-(A\cup B))\cap A\}-(A-B)$
$=\{(U\cap A)-\{(A\cup B)\cap A\}\}-(A-B)$
$=\{A-A\}-(A-B)=\phi-(A-B)=\phi$

35. *(a)* $\{a\}$ is an element of $\{\{a\},\{b\},\{c\}\}$, subsets of $\{\{a\}, b, c\}$ are ϕ, $\{\{a\}$, $\{b\}$, $\{c\}$, $\{\{a\}, b\}$, $\{b, c\}$, $\{\{a\}, c\}$, $\{\{a\}, b, c\}$
Hence, only statement I is correct.

36. *(d)* $\{x: x$ is an integer and less than $1000\}=[\dots,998,999]$
i.e. $\;x\in(-\infty, 1000)$ is an infinite set.

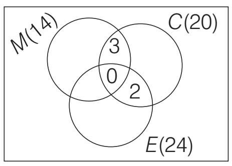
37. *(d)* If $A\cap B=\phi$, then it is not necessary that either.
$A=\phi$ or $B=\phi$.

38. *(b)* Since $R\in\overleftrightarrow{PQ}$ i.e R belongs to $\overleftrightarrow{PQ}$.
Again $S,\overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended
$\therefore \quad \overleftrightarrow{PQ}\cap\overleftrightarrow{RS}=\{R\}$
where R is point of intersection of the straight lines PQ and RS.

39. *(a)* Here, H = People, who can speak Hindi
E = People, who can speak English
Given, $n(H\cup E)=1000, n(H)=750$,
$n(E)=400$
$n(H\cup E)=n(H)+n(E)-n(H\cap E)$
$1000=750+400-n(H\cap E)$
$\Rightarrow n(H\cap E)=1150-1000=150$
Number of people who can speak Hindi only
$=n(H)-n(H\cap E)$
$=750-150=600$

40. *(a)* Let S be the set of people who speak spanish and F be the set of people who speak French
$\therefore \quad n(S)=20, n(F)=50, n(S\cap F)=10$
$\Rightarrow \quad n(S\cup F)=n(S)+n(F)-n(S\cap F)$
$=20+50-10=60$

41. *(d)* Here, M = Students who study Mathematics
E = Students who study English
C = Students who study Commerce
$\therefore n(M\cup E\cup C)=42$
Also, $n(M)=14, n(E)=24, n(C)=20$,
$n(M\cap C)=3$, $n(E\cap C)=2$
and $n(M\cap E\cap C)=0$



From the Venn diagram, it is clear that number of students who study Mathematics but not Commerce
$=14-3=11$

42. *(c)* Let number of failed candidates in Mathematics and English be M and E, respectively.
Given, candidates failed in English, $n(E)=52\%$,
Candidates failed in Mathematics, $n(M)=42\%$
Candidates failed in both English and Mathematics,
$n(E\cap M)=17\%$
$\because$ Total candidates, failed
$n(E\cup M)=n(E)+n(M)-n(E\cap M)$
$=52+42-17=77\%$
$\therefore$ Total candidates passed $=(100-77)$
$=23\%$

43. *(c)* Clearly, both I and II are true.

44. *(c)* I. $\quad A=\{0\}\qquad$ II. $B=\{2\}$
III. $C=\{\ \}$; ± 4 is not an odd integer
Here, only III is empty set.

45. *(c)* By distributive law in sets
$A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$
So, only III is correct.

46. *(c)* Here, $P=\{\dots-6,-3,0,3,6,\dots\}$,
$Q=\{\dots-8,-4,0,4,8,\dots\}$ and
$R=\{\dots-12,-6,0,6,12,\dots\}$
I. $P\cup Q=\{\dots-8,-6,-4,-3,0,3,4,6,8,\dots\}\neq R$
II. $P\not\subset R$ as $3\in P$ but $3\notin R$.
III. $R\subset(P\cup Q)$ is true.
Hence, only statement III is correct.

47. *(a)* I. There are infinite points lie on a line segment so, it is an infinite set.
II. Number of birds in zoo are countable so, it is a finite set.
III. It is not a well-defined set.
Hence, only I is correct.

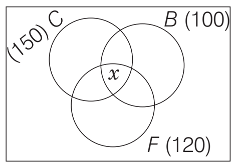
48. *(a)* I. A is a finite set.
II. As all elements of A are integers, so A is subset of integers.
III. $\{1, 2\}$ is a proper subset of A.
IV. $A\neq\phi$.
So, I, II and III are true.

49. *(d)* All sets are infinite set.

50. *(c)* Equivalent sets have same cardinal numbers. Here, cardinal numbers of I, III, IV sets are same.

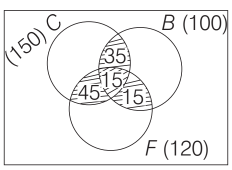
Solutions (Q. No. 51-52) *Let C, B and F denotes the number of students who play cricket, basketball and football respectively.*

Given, $n(C)=150$, $n(B)=100$, $n(F)=120$, $n(B\cap F)=30$, $n(C\cap B)=50$ and $n(C\cap F)=60$
Let x be the number of students who play all the three sports



51. *(a)* $n(C \cup B \cup F) = n(C) + n(B)$
$+ n(F) - n(C \cap F)$
$- n(C \cap B) - n(B \cap F) + n(C \cap B \cap F)$
$\Rightarrow \quad n(C \cup B \cup F) = 150 + 100$
$+ 120 - 60 - 50 - 30 + x$
$\Rightarrow \quad n(C \cup B \cup F) = 230 + x$
Since, maximum value of $n(C \cup B \cup F)$ is 250
Hence, maximum value of
$x = 250 - 230 = 20$

52. *(b)* Here, $n(C \cup B \cup F) = 250 - 5 = 245$
$\therefore \quad 245 = 230 + x \quad \Rightarrow \quad x = 15$



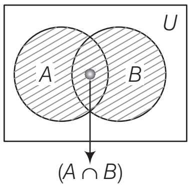
Number of students who play atleast two sports $= 15 + 35 + 45 + 15 = 110$

53. *(a)* Given, $A = \{2, 4, 6, 8, 10, 12, \ldots\}$,
$B = \{5, 10, 15, 20, 25, 30, \ldots\}$
and $\quad C = \{10, 20, 30, 40, 50, 60, \ldots\}$
Now,
$(B \cup C) = \{5, 10, 15, 20, 25, 30, \ldots\}$
$\therefore \; A \cap (B \cup C) = \{2, 4, 6, 8, 10, 12, \ldots\}$
$\cap \{5, 10, 15, 20, 25, 30, \ldots\}$
$= \{10, 20, 30, \ldots\}$

54. *(a)* From option (a), $A = \{x$ is a real number : $x > 1$ and $x < 1\}$.
Since, there is no such element which is greater than 1 and less than 1.
So, A is a null set.
From option (b), $B = \{x : x + 3 = 3\} = \{0\}$ = Singleton set
From option (c), $C = \{\phi\}$ = Singleton set
From option (d), $D = \{x$ is a real number : $x \geq 1$ and $x \leq 1\}$
$= \{1\}$ = Singleton set

55. *(d)* Given that, $x \in \{2, 3, 4\}$
and $y \in \{4, 6, 9, 10\}$
and also $A = (x, y)$, such that x is a factor of y.
$\therefore \quad A = \{(2, 4), (2, 6), (2, 10), (3, 6), (3, 9), (4, 4)\}$
Hence, A contain 6 elements.

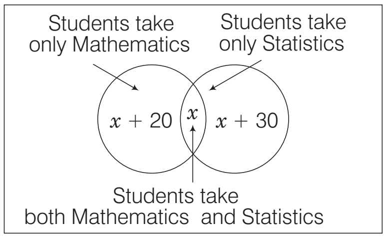
56. *(d)* From figure,



I. $(A \cap B) \subseteq A$ [true]
II. $(A \cap B) \subseteq B$ [true]
III. $A \subseteq (A \cup B)$ [true]
So, all three statements are correct.
Shaded region $= (A \cup B)$

57. *(c)* Let M and P be the teachers who teach Mathematics and Physics, respectively.
Given,
$n(M \cup P) = 30$, $n(M) = 20$, $n(P) = 15$ and $n(M \cap P) = 5$
$\therefore$ Number of teachers teaching only Mathematics is
$n(\text{only } M) = n(M) - n(M \cap P)$
$= 20 - 5 = 15$

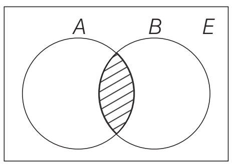
58. *(b)* Venn diagram of given conditions is as shown in figure given below



Total students = Students those take only Mathematics + Student those take only Statistics + Student those take both Mathematics and Statistics
$\because \quad 110 = x + 20 + x + x + 30$
$\Rightarrow \quad 110 = 3x + 50 \quad \Rightarrow 3x = 60$
$\therefore \quad x = 20$

59. *(b)* $E \cup \{(A \cap \phi) - (A - \phi)\}$
$= E \cup \phi - A = E - A = A'$
$[\because (A \cap \phi) = \phi \;\text{ and }\; (A - \phi) = A]$

60. *(c)* Since, A and B are non-empty subsets of E.



$\therefore A \cup (A \cap B) = A \cup$ (Shaded portion)
$= A$
If A and B are two non-empty disjoint sets.
then, $\quad A \cap B = \phi$
$\therefore \quad A \cup (A \cap B) = A \cup \phi = A$

61. *(c)* Let E and M represent the students pass in English and Mathematics.
Given, $\quad n(U) = 105$, $n(E) = 80$,
$n(M) = 75$, and $n(E \cap M) = 10$
Now, $n(E \cup M) = n(U) - n(E \cap M)$
$= 105 - 10 = 95$
Also, $n(E \cup M) = n(E) + n(M)$
$- n(E \cap M)$
$\Rightarrow \quad 95 = 80 + 75 - n(E \cap M)$
$\Rightarrow \quad n(E \cap M) = 60$
The number of students who pass in only one subject
$= n(E) + n(M) - 2\,n(E \cap M)$
$= 80 + 75 - 2 \times 60$
$= 155 - 120 = 35$

62. *(c)* A = diagonal equal and bisecting each other.
A is square or rectangle. and B diagonal bisecting each other at 90°.
So, $A \cap B$ = the set of squares.

63. *(d)* The possible set of pairs *(a, b)* such that *ab* leaves remainder 1 when divided by 15 are (2, 8), (8, 2) (7, 13) and (13, 7).
$\therefore$ Number of possible set of pairs = 4

64. *(a)* Let A, B and C be the number of people who can speak Hindi, English and French.
Then, $n(A) = 70$, $n(B) = 60$,
$n(C) = 30$, $n(A \cap B) = 30$,
$n(A \cap C) = 20$, and $n(B \cap C) = x$
$n(A \cup B \cup C) = n(A) + n(B) + n(C)$
$- n(A \cap B) - n(B \cap C)$
$- n(A \cap C) + n(A \cap B \cap C)$
$\Rightarrow \quad 100 = 70 + 60 + 30 - 30 - 20$
$- x + n(A \cap B \cap C)$
$\Rightarrow \quad x = 10 + n(A \cap B \cap C)$
Also, $n(A \cap B \cap C)$ can't be more than $n(A \cap C)$
$\Rightarrow \quad 10 \leq x \leq 30$

20

MEASUREMENTS OF ANGLES AND TRIGONOMETRIC RATIOS

Generally (13-14) questions have been asked from this chapter. Generally questions asked from this chapter are based on the trigonometric identities and formulae. So, detailed study of this chapter will help you to score in exam as good number of questions have been asked from this chapter.

Trigonometry is the branch of Mathematics which deals with the measurements of sides and angles of triangles and the problems based on these angles.

ANGLE

An angle is considered as the figure obtained by rotating a given ray about its endpoint. The original ray is called the initial side and the ray into which the initial side rotates is called terminal side. In figure, OA is initial side and OB is the terminal side of $\angle AOB$. The point O is called its vertex.

QUADRANTS

Let XOX' and YOY' be two lines at right angles in the plane of paper. These two perpendicular lines divide the plane of the paper into four equal parts, these four parts are known as four quadrants and lines XOX' and YOY' are known as X-axis and Y-axis, respectively.

The parts $XOY, YOX', X'OY'$ and $Y'OX$ are known as 1st, 2nd, 3rd and 4th quadrant, respectively.

Relation between Angles, Radius and Arc Length

If l' is the length of a circle of radius r. And this arc subtends an angle θ radians at the centre of the circle, then $l = r\theta \Rightarrow \theta = \frac{l}{r}$

Relation between Degree and Radian

A circle subtends an angle, at the centre whose radian measure is 2π and its degree measure is $360°$. It follows that

$$2\pi \text{ radian} = 360°$$
$$\pi \text{ radian} = 180°$$
$$\therefore \quad 1 \text{ radian} = \frac{180}{\pi} \text{ degree} = 57° \, 16' \, 22'' \quad \text{(approx)}$$
$$1 \text{ degree} = \frac{\pi}{180} \text{ radian} = 0.01746 \text{ rad} \quad \text{(approx)}$$

Note 1 radian is written as 1^c.

EXAMPLE 1. The radian measure corresponding to $-22° \, 30'$ is

a. $\frac{\pi^c}{8}$ b. $-\frac{\pi^c}{8}$ c. $\frac{\pi^c}{4}$ d. $-\frac{\pi^c}{4}$

Sol. b. As, $180° = \pi$

$$\therefore -22°30' = -\left(22 + \frac{1}{2}\right) = -\left(\frac{45}{2}\right)^{\circ} = -\left(\frac{45}{2} \times \frac{\pi}{180}\right)^{c} = -\frac{\pi^c}{8}$$

IMPORTANT POINTS

- The angle between two consecutive digits in a clock is $30° = \left(\frac{\pi}{6} \text{radian}\right)$.
- The hour hand rotates through an angle of $30°$ in one hour i.e. $\left(\frac{1}{2}\right)^{\circ}$ in one minute.
- The minute hand rotates through an angle of $6°$ in one minute.

EXAMPLE 2. If an arc 24π cm long of a circle subtends an angle of $72°$ at its centre, then the radius of the circle is

a. 30 cm b. 40 cm c. 50 cm d. 60 cm

Sol. d. Given, length of arc $(l) = (24\pi)$ cm

$$\text{Angle } (\theta) = 72° = \left(72 \times \frac{\pi}{180}\right)^{c} = \left(\frac{2\pi}{5}\right)^{c}$$

Let r be the radius of circle

$$\text{Since, } \theta = \frac{l}{r} \Rightarrow r = \frac{l}{\theta} = \frac{24\pi}{\frac{2\pi}{5}} = 60 \text{ cm}$$

EXAMPLE 3. Consider the following statements:

I. The angular measure in radian of a circular arc of fixed length subtending at its centre decreases, if the radius of the arc increases.

II. $1800°$ is equal to 5π radian.

Which of the statements given above is/are correct? **2012 I**

a. Only I b. Only II
c. Both I and II d. Neither I nor II

Sol. a. I. We know that,

$$\text{Radius} = \frac{\text{arc}}{\text{angle}} \qquad \text{(given, arc length is constant)}$$
$$\therefore \text{Radius} \propto \frac{1}{\text{angle}}$$

So, angular measure in radian decreases, if the radius of the arc increases.

II. $1800° \times \frac{\pi}{180°} = 10\pi$

Hence, only statement I is correct.

EXAMPLE 4. The angle between the hour hand and the minute hand of a clock at half-past three is

a. 54° b. 63° c. 75° d. 85°

Sol. c. Angle traced by hour hand in 1 hr $= 30°$

Angle traced by hour hand in $3\frac{1}{2}$ hr $= \left(30 \times \frac{7}{2}\right)^{\circ} = 105°$

Angle traced by minute hand in 1 min $= 6°$

Angle traced by minute hand in 30 min $= 6 \times 30° = 180°$

Angle between two hands $= 180° - 105° = 75°$

TRIGONOMETRIC RATIOS

The ratio of the sides of a right angled triangle with respect to its angle are called trigonometric ratios.

The side opposite to the right angle is called the hypotenuse. Relative to the angle θ, the side opposite to angle θ is caused the prependicular side and the remaining side is called base.

(i) For angle θ, perpendicular $= BC$, base $= AB$ and hypotenuse $= AC$

(ii) For angle α, perpendicular $= AB$, base $= BC$ and hypotenuse $= AC$

Trigonometric ratios of θ in right angled ΔABC are defined below

$$\sin\theta = \frac{Perpendicular}{Hypotenuse} = \frac{BC}{AC} = \frac{P}{H}$$
$$\cos\theta = \frac{Base}{Hypotenuse} = \frac{AB}{AC} = \frac{B}{H}$$
$$\tan\theta = \frac{Perpendicular}{Base} = \frac{BC}{AB} = \frac{P}{B}$$
$$\sec\theta = \frac{Hypotenuse}{Base} = \frac{AC}{AB} = \frac{H}{B}$$
$$\cot\theta = \frac{Base}{Perpendicular} = \frac{AB}{BC} = \frac{B}{P}$$
$$cosec\,\theta = \frac{Hypotenuse}{Perpendicular} = \frac{AC}{BC} = \frac{H}{P}$$